

P₁ 习题8 P₂₁ 2, 3

$$II). \frac{\partial z}{\partial t} = \frac{\partial z}{\partial x} \frac{dx}{dt} + \frac{\partial z}{\partial y} \frac{dy}{dt} = -\frac{1}{\sqrt{1-(x-y)^2}} \times 3 + \frac{1}{\sqrt{1-(x-y)^2}} \times 8t = \frac{8t-3}{\sqrt{1-(x-y)^2}}$$

$$III) u = u(x, y, z)$$

$$\begin{aligned} \frac{du}{dx} &= \frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} \frac{dy}{dx} + \frac{\partial u}{\partial z} \frac{dz}{dx} \\ &= \frac{a(y-z)}{a^2+1} e^{ax} + \frac{e^{ax}}{a^2+1} \cdot a \cos x + \frac{e^{ax}}{a^2+1} \cdot \sin x \\ &= e^{ax} \left(\frac{a(a \sin x - \cos x)}{a^2+1} + \frac{a \cos x}{a^2+1} + \frac{\sin x}{a^2+1} \right) = e^{ax} \sin x \end{aligned}$$

P₂ 习题8 P₂₂ 2, 4

$$II) \quad \begin{cases} u = x^2 y^2 \\ v = xy \\ z = u e^{\frac{u}{v}} \end{cases}$$

$$\frac{\partial z}{\partial x} = \frac{\partial z}{\partial u} \frac{\partial u}{\partial x} + \frac{\partial z}{\partial v} \frac{\partial v}{\partial x} = (e^{\frac{u}{v}} + \frac{u}{v} e^{\frac{u}{v}}) \cdot 2x - \frac{u^2}{v^2} e^{\frac{u}{v}} \cdot y = e^{\frac{xy^2}{xy}} (2x + \frac{x^2 y^2}{xy})$$

$$\frac{\partial z}{\partial y} = \frac{\partial z}{\partial u} \frac{\partial u}{\partial y} + \frac{\partial z}{\partial v} \frac{\partial v}{\partial y} = e^{\frac{xy^2}{xy}} (2y + \frac{y^4 - x^2}{y^2 x})$$

$$IV) \quad \begin{cases} z = x^{xy} \\ \ln z = xy \ln x \end{cases}$$

$$\frac{\partial z}{\partial x} = y x^{y-1} \ln x + x^{y-1} \rightarrow \frac{\partial z}{\partial x} = x^{x+y-1} (y \ln x + 1)$$

$$\frac{\partial z}{\partial y} = (\ln^2 x) \cdot x^y \rightarrow \frac{\partial z}{\partial y} = x^{x+y} (\ln^2 x)$$

P₃ 习题8 P₂₃ 3, 4

$$III). \quad \begin{cases} u = \frac{y}{x} \\ z = y f(u) \end{cases}$$

$$\frac{\partial z}{\partial x} = -\frac{y^2}{x^2} f'(\frac{y}{x})$$

$$\frac{\partial z}{\partial y} = f(\frac{y}{x}) + \frac{y}{x} f'(\frac{y}{x})$$

$$IV) z = f(t, ts, tsr)$$

$$\frac{\partial z}{\partial t} = f_1 + sf_2 + srf_3$$

$$\frac{\partial z}{\partial s} = tf_2 + trf_3$$

$$\frac{\partial z}{\partial r} = tsf_3$$

P₄ 题8 P₂₅

$$\begin{cases} x = r \cos \theta \\ y = r \sin \theta \end{cases} \quad \begin{aligned} \frac{\partial u}{\partial r} &= \frac{\partial u}{\partial x} \cos \theta + \frac{\partial u}{\partial y} \sin \theta \\ \frac{\partial u}{\partial \theta} &= -\frac{\partial u}{\partial x} r \sin \theta + \frac{\partial u}{\partial y} r \cos \theta \\ \frac{\partial v}{\partial r} &= \frac{\partial v}{\partial x} \cos \theta + \frac{\partial v}{\partial y} \sin \theta \\ \frac{\partial v}{\partial \theta} &= -\frac{\partial v}{\partial x} r \sin \theta + \frac{\partial v}{\partial y} r \cos \theta \end{aligned}$$

证代:

$$\frac{1}{r} \frac{\partial u}{\partial \theta} = \frac{1}{r} \left(-\frac{\partial u}{\partial x} r \sin \theta + \frac{\partial u}{\partial y} r \cos \theta \right) = -\frac{\partial u}{\partial x} \sin \theta + \frac{\partial u}{\partial y} \cos \theta = \frac{\partial u}{\partial y} \sin \theta + \frac{\partial u}{\partial x} \cos \theta = \frac{\partial u}{\partial r} \rightarrow \frac{\partial u}{\partial r} = \frac{1}{r} \frac{\partial u}{\partial \theta}$$

$$-\frac{1}{r} \frac{\partial v}{\partial \theta} = -\frac{1}{r} \left(-\frac{\partial v}{\partial x} r \sin \theta + \frac{\partial v}{\partial y} r \cos \theta \right) = \frac{\partial v}{\partial x} \sin \theta - \frac{\partial v}{\partial y} \cos \theta = \frac{\partial v}{\partial y} \sin \theta + \frac{\partial v}{\partial x} \cos \theta = \frac{\partial v}{\partial r} \rightarrow \frac{\partial v}{\partial r} = -\frac{1}{r} \frac{\partial v}{\partial \theta} \quad \square$$

P₅ 题8 P₂₆

$$I) \frac{\partial z}{\partial x} = \frac{\partial z}{\partial u} \frac{\partial u}{\partial x} + \frac{\partial z}{\partial v} \frac{\partial v}{\partial x} = \frac{x}{x^2+y^2} \frac{\partial z}{\partial u} - \frac{y}{x^2+y^2} \frac{\partial z}{\partial v}$$

$$\frac{\partial z}{\partial y} = \frac{\partial z}{\partial u} \frac{\partial u}{\partial y} + \frac{\partial z}{\partial v} \frac{\partial v}{\partial y} = \frac{y}{x^2+y^2} \frac{\partial z}{\partial u} + \frac{x}{x^2+y^2} \frac{\partial z}{\partial v}$$

$$\text{证代: } \frac{x(x+y) - y(x-y)}{x^2+y^2} \frac{\partial z}{\partial u} - \frac{y(x+y) + x(x-y)}{x^2+y^2} \frac{\partial z}{\partial v} = 0$$

$$\text{整理得 } \frac{\partial z}{\partial u} - \frac{\partial z}{\partial v} = 0$$

$$II) z = e^{u+xy}$$

$$\frac{\partial z}{\partial x} = e^{u+xy} \left(\frac{\partial u}{\partial x} + 1 \right) \quad \frac{\partial z}{\partial y} = e^{u+xy} \left(\frac{\partial u}{\partial y} + x \right)$$

$$y z \left(\frac{\partial u}{\partial x} + 1 \right) - x z \left(\frac{\partial u}{\partial y} + x \right) = y - x^2 z$$

$$y \frac{\partial u}{\partial x} - x \frac{\partial u}{\partial y} = 0$$

$$\frac{\partial w}{\partial x} = \frac{\partial w}{\partial u} \cdot 2x - \frac{\partial w}{\partial v} \cdot \frac{1}{x^2} \quad \frac{\partial w}{\partial y} = \frac{\partial w}{\partial u} \cdot 2y - \frac{\partial w}{\partial v} \cdot \frac{1}{y^2}$$

$$2xy \frac{\partial w}{\partial u} - \frac{\partial w}{\partial v} \frac{y}{x^2} = 2xy \frac{\partial w}{\partial u} - \frac{\partial w}{\partial v} \frac{x}{y^2}$$

$$\frac{\partial w}{\partial v} \left(\frac{y}{x^2} - \frac{x}{y^2} \right) = 0$$

$$\frac{\partial w}{\partial v} = 0$$

P6 习题 8 P7 1.4

$$I) z = f(xy^2, xy)$$

$$\frac{\partial z}{\partial x} = y^2 f_1 + 2xy f_2 \quad \frac{\partial z}{\partial y} = 2xy f_1 + x^2 f_2$$

$$\frac{\partial^2 z}{\partial x^2} = y^2 (y^2 f_{11} + 2xy f_{12}) + 2y f_2 + 2xy (y^2 f_{21} + 2xy f_{22}) = 2y f_2 + y^4 f_{11} + 4xy^3 f_{12} + 4xy^2 f_{22}$$

$$\frac{\partial^2 z}{\partial x \partial y} = 2y f_1 + y^2 (2xy f_{11} + x^2 f_{12}) + 2x f_2 + 2xy (2xy f_{21} + x^2 f_{22}) = 2(y f_1 + x f_2) + 2xy^3 f_{11} + 2x^3 y f_{22} + 5x^2 y^2 f_{12}$$

$$\frac{\partial^2 z}{\partial y^2} = 2x f_1 + 2xy (2xy f_{11} + x^2 f_{12}) + x (2xy f_{21} + x^2 f_{22}) = 2x f_1 + 4x^2 y^2 f_{11} + 4x^3 y f_{12} + x^4 f_{22}$$

$$II) z = f(x+y, xy, \frac{y}{x})$$

$$\frac{\partial z}{\partial x} = f_1 + y f_2 + \frac{1}{y} f_3 \quad \frac{\partial z}{\partial y} = f_1 + x f_2 - \frac{x}{y^2} f_3$$

$$\frac{\partial^2 z}{\partial x^2} = (f_{11} + y f_{12} + \frac{1}{y} f_{13}) + y (f_{21} + y f_{22} + \frac{1}{y} f_{23}) + \frac{1}{y} (f_{31} + y f_{32} + \frac{1}{y} f_{33}) = f_{11} + y^2 f_{22} + \frac{1}{y^2} f_{33} + 2y f_{12} + \frac{2}{y} f_{13} + 2f_{23}$$

$$\frac{\partial^2 z}{\partial x \partial y} = (f_{11} + x f_{12} - \frac{x}{y^2} f_{13}) + f_2 + y (f_{21} + x f_{22} - \frac{x}{y^2} f_{23}) - \frac{x}{y^2} f_3 + \frac{1}{y} (f_{31} + x f_{32} - \frac{x}{y^2} f_{33}) = f_2 - \frac{x}{y^2} f_3 + f_{11} + xy f_{22} - \frac{x}{y^2} f_{33} + (x+y) f_{12} + (\frac{1}{y} - \frac{x}{y^2}) f_{13}$$

$$\frac{\partial^2 z}{\partial y^2} = (f_{11} + x f_{12} - \frac{x}{y^2} f_{13}) + x (f_{21} + x f_{22} - \frac{x}{y^2} f_{23}) + \frac{2x}{y^2} f_3 - \frac{x}{y^2} (f_{31} + x f_{32} - \frac{x}{y^2} f_{33}) = \frac{2x}{y^2} f_3 + f_{11} + x^2 f_{22} + \frac{x^2}{y^2} f_{33} + 2x f_{12} - \frac{2x}{y^2} f_{13} - \frac{2x^2}{y^2} f_{23}$$

P7 习题 8 P28

$$I) F(x, y) = \int_1^x \left[\int_0^y f(t, u) g\left(\frac{t}{u}\right) dt \right] du$$

$$\frac{\partial F}{\partial x} = \int_0^y f(x, u) g\left(\frac{x}{u}\right) du$$

$$\frac{\partial^2 F}{\partial x \partial y} = x f(x, y) g\left(\frac{x}{y}\right)$$

$$II) F(x, y) = \int_0^{\pi y} f(t, e^t) dt$$

$$\frac{\partial F}{\partial x} = 2\pi y f(\pi y, e^{\pi y})$$

$$\frac{\partial F}{\partial x \partial y} = 2\pi f + 2\pi y (\pi^2 f_1 + \pi^2 e^{\pi y} f_2) = 2\pi f + 2\pi y (\pi f_1 + e^{\pi y} f_2)$$

P8 习题8 P30

$$\frac{\partial z}{\partial x} = e^x \sin y f'(u) \quad \frac{\partial z}{\partial y} = e^x \cos y f'(u)$$

$$\frac{\partial^2 z}{\partial x^2} = e^x \sin y f''(u) + (e^x \sin y)^2 f''(u) \quad \frac{\partial^2 z}{\partial y^2} = -e^x \sin y f''(u) + (e^x \cos y)^2 f''(u)$$

$$\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = e^{2x} f''(u) = e^{2x} f''(u)$$

$$f''(u) - f''(u) = 0$$

$$r^2 - 1 = 0 \rightarrow r_1 = 1, r_2 = -1 \rightarrow e^x, e^{-x}$$

$$f(u) = c_1 e^x + c_2 e^{-x}$$

P9 习题8 P32

$$\frac{\partial z}{\partial x} = \frac{\partial z}{\partial u} + \frac{\partial z}{\partial v} \quad \frac{\partial z}{\partial y} = a \frac{\partial z}{\partial u} + b \frac{\partial z}{\partial v}$$

$$z_{xx} = z_{uu} + 2z_{uv} + z_{vv} \quad z_{xy} = a z_{uu} + (a+b) z_{uv} + b z_{vv} \quad z_{yy} = a^2 z_{uu} + 2ab z_{uv} + b^2 z_{vv}$$

$$\text{代入: } (3a^2 + 4a + 1) z_{uu} + (b^2 + 4a + 4b + 2) z_{uv} + (3b^2 + 4b + 1) z_{vv} = 0$$

$$\begin{cases} 3a^2 + 4a + 1 = 0 \\ b^2 + 4a + 4b + 2 \neq 0 \\ 3b^2 + 4b + 1 = 0 \end{cases} \xrightarrow{a \neq b} \begin{cases} a = -1 \\ b = -\frac{1}{3} \end{cases} \text{ 或 } \begin{cases} a = -\frac{1}{3} \\ b = -1 \end{cases}$$

P10 习题8 P33 1, 4

$$I) \frac{\partial F}{\partial x} = y \cos \pi y - y e^{\pi y} - 2\pi y \quad \frac{\partial F}{\partial y} = x \cos \pi y - x e^{\pi y} - x^2$$

$$\frac{dy}{dx} = \frac{-F_x}{F_y} = -\frac{y}{x} \frac{\cos \pi y - e^{\pi y} - 2\pi}{\cos \pi y - e^{\pi y} - x^2}$$

$$IV) F(x, y) = \sin \pi y - \ln(x+1) + \ln y - 1$$

$$\frac{\partial F}{\partial x} = y \cos \pi y - \frac{1}{x+1} \quad \frac{\partial F}{\partial y} = \pi \cos \pi y + \frac{1}{y}$$

$$\frac{dy}{dx} = -\frac{F_x}{F_y} = -\frac{y \cos \pi y - \frac{1}{x+1}}{\pi \cos \pi y + \frac{1}{y}}$$

$$x=0 \text{ 时, } \ln \frac{1}{y} + 1 = 0 \rightarrow y = e$$

$$y'(0) = -\frac{y-1}{\frac{1}{y}} = -e(e-1) = e - e^2$$

P₁₁ 习题 8 P₃₄ 1.4

$$I) F(x, y, z) = \ln \sin \frac{z}{y} - \frac{x}{z}$$

$$F_x = -\frac{1}{z} \quad F_y = -\frac{1}{y^2} \cot \frac{z}{y} \quad F_z = \frac{1}{y} \cot \frac{z}{y} + \frac{1}{z^2}$$

$$\frac{\partial z}{\partial x} = -\frac{F_x}{F_z} = \frac{\frac{1}{z}}{\frac{1}{y} \cot \frac{z}{y} + \frac{1}{z^2}} = \frac{yz}{z^2 \cot \frac{z}{y} + \pi y}$$

$$\frac{\partial z}{\partial y} = -\frac{F_y}{F_z} = \frac{\frac{1}{y^2} \cot \frac{z}{y}}{\frac{1}{y} \cot \frac{z}{y} + \frac{1}{z^2}} = \frac{z^2 \cot \frac{z}{y}}{yz^2 \cot \frac{z}{y} + \pi y z}$$

$$IV) F(x, y, z) = z^2 - 2z + \pi^2 y^2$$

$$F_x = 2x \quad F_y = 2y \quad F_z = 2z - 2$$

$$dz = \frac{\partial z}{\partial x} dx + \frac{\partial z}{\partial y} dy = \frac{x}{1-z} dx + \frac{y}{1-z} dy = \frac{x dx + y dy}{1-z}$$

P₁₂ 习题 8 P₃₅ 3

$$\frac{1}{2} F(x, y) = e^{\pi y} - \pi y - 2$$

$$F_x = ye^{\pi y} - y \quad F_y = \pi e^{\pi y} - \pi \rightarrow \frac{dy}{dx} = -\frac{F_x}{F_y} = -\frac{y}{x}$$

$$\frac{1}{2} G(x, z) = e^x - \int_0^{\pi z} \frac{\sin t}{t} dt$$

$$G_x = e^x - \frac{\sin(\pi z)}{\pi z} \quad G_z = \frac{\sin(\pi z)}{\pi z} \rightarrow \frac{dz}{dx} = -\frac{G_x}{G_z} = \frac{\sin(\pi z) - e^{\pi(\pi z)}}{\sin(\pi z)}$$

$$\frac{du}{dx} = f_x - \frac{y}{x} f_y + \frac{\sin(\pi z) - e^{\pi(\pi z)}}{\sin(\pi z)} f_z$$

P13 习题8 P36 2.4

II) 令 $F(x, y, z) = x + y\varphi(z) - z$

$$F_x = 1 \quad F_y = \varphi(z) \quad F_z = y\varphi'(z) - 1$$

$$\frac{\partial u}{\partial y} = f(z) \frac{\partial z}{\partial y} = f(z) \frac{\varphi'(z)}{1 - y\varphi'(z)}$$

$$\frac{\partial u}{\partial x} = f(z) \frac{\partial z}{\partial x} = f(z) \frac{1}{1 - y\varphi'(z)}$$

故 $\frac{\partial u}{\partial y} = \varphi(z) \frac{\partial u}{\partial x} \quad \square$

IV) 令 $F(x, y, z) = \varphi(x^2 + y^2 + z^2) - ax - by - cz$

$$F_x = 2x\varphi'(x^2 + y^2 + z^2) - a \quad F_y = 2y\varphi'(x^2 + y^2 + z^2) - b \quad F_z = 2z\varphi'(x^2 + y^2 + z^2) - c$$

$$\frac{\partial z}{\partial x} = -\frac{2x\varphi'(x^2 + y^2 + z^2) - a}{2z\varphi'(x^2 + y^2 + z^2) - c} \quad \frac{\partial z}{\partial y} = -\frac{2y\varphi'(x^2 + y^2 + z^2) - b}{2z\varphi'(x^2 + y^2 + z^2) - c}$$

$$(cy - bz) \frac{\partial z}{\partial x} + (az - cx) \frac{\partial z}{\partial y}$$

$$= -\frac{(ay - bx)(2z\varphi'(x^2 + y^2 + z^2) - c)}{2z\varphi'(x^2 + y^2 + z^2) - c} = bx - ay \quad \square$$

P15 习题8 P39 1.4

I) $\begin{cases} z - x^2 - y^2 = 0 \\ x^2 + 2y^2 + 3z^2 - 4 = 0 \end{cases} \rightarrow \begin{cases} \frac{dz}{dx} - 2x - 2y \frac{dy}{dx} = 0 \\ 2x + 4y \frac{dy}{dx} + 6z \frac{dz}{dx} = 0 \end{cases}$

$$J = \begin{vmatrix} -2y & 1 \\ 4y & 3z \end{vmatrix} = -2y(3z+1)$$

$$\frac{dy}{dx} = \frac{\begin{vmatrix} 2x & 1 \\ -x & 3z \end{vmatrix}}{-2y(3z+1)} = -\frac{x(3z+1)}{2y(3z+1)}$$

$$\frac{dz}{dx} = \frac{\begin{vmatrix} -2y & 2x \\ 4y & -x \end{vmatrix}}{-2y(3z+1)} = \frac{x}{3z+1} \quad (y \neq 0, 3z+1 \neq 0)$$

IV). $\begin{cases} x = e^u \cos v \\ y = e^u \sin v \\ z = uv \end{cases}$

求全微分: $\begin{cases} dx = e^u \cos v du - e^u \sin v dv = x du - y dv \\ dy = e^u \sin v du + e^u \cos v dv = y du + x dv \\ dz = v du + u dv \end{cases} \rightarrow \begin{cases} du = \frac{x dx + y dy}{x^2 + y^2} \\ dv = \frac{-y dx + x dy}{x^2 + y^2} \end{cases}$

$$dz = \frac{vx - uy}{x^2 + y^2} dx + \frac{vy + ux}{x^2 + y^2} dy \rightarrow \begin{cases} \frac{\partial z}{\partial x} = \frac{vx - uy}{x^2 + y^2} \\ \frac{\partial z}{\partial y} = \frac{vy + ux}{x^2 + y^2} \end{cases}$$

P₁₆ 习题8 P₄₀ 2

$$u_x = \frac{x}{\sqrt{x^2+y^2+z^2}} \quad u_y = \frac{y}{\sqrt{x^2+y^2+z^2}} \quad u_z = \frac{z}{\sqrt{x^2+y^2+z^2}}$$

$$\cos \alpha = \frac{1}{3} \quad \cos \beta = \frac{2}{3} \quad \cos \gamma = \frac{2}{3}$$

$$\frac{\partial u}{\partial r} \Big|_{m0} = \frac{\sqrt{2}}{2} \times \frac{1}{3} + \frac{\sqrt{2}}{2} \times \frac{2}{3} = \frac{\sqrt{2}}{2}$$

P₁₇ 习题8 P₄₁ 2, 3

$$\text{II) } u = 3x^2 - 2y^2 + 3z^2$$

$$\nabla u = (6x, -4y, 6z)$$

$$\text{III) } \nabla u = \left(\frac{xz^2}{\sqrt{x^2+2y^2}}, \frac{2yz^2}{\sqrt{x^2+2y^2}}, 2z\sqrt{x^2+2y^2} \right)$$

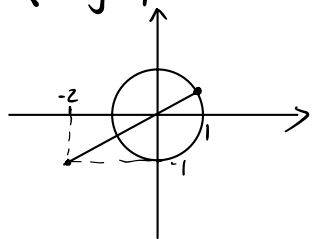
$$\nabla u \Big|_{(1, \frac{\sqrt{2}}{2}, 1)} = \left(\frac{\sqrt{2}}{2}, 1, 2\sqrt{2} \right)$$

P₁₈ 补充8 P₅

$$\frac{\partial f}{\partial x} = 1+y \quad \frac{\partial f}{\partial y} = 2+x$$

$$\nabla f = (1+y, 2+x) \quad |\nabla f|^2 = (1+y)^2 + (2+x)^2$$

$$\text{有 } x^2+y^2=1$$



$$\nabla f_{\max} = \sqrt{5} + 1$$

P₁₉ 补充8 P₆

$$\begin{cases} x=f(u,v) \\ y=g(u,v) \\ z=h(u,v) \end{cases} \text{ 求全微分 } \begin{cases} dx=f_u du+f_v dv \\ dy=g_u du+g_v dv \\ dz=h_u du+h_v dv \end{cases} \rightarrow \begin{cases} du = \frac{g_v dx - f_v dy}{f_u g_v - f_v g_u} \\ dv = \frac{f_u dy - g_u dx}{f_u g_v - f_v g_u} \end{cases}$$

$$\text{代入: } dz = \frac{h_u h_v - h_v h_u}{f_u g_v - f_v g_u} dx + \frac{h_v f_u - h_u f_v}{f_u g_v - f_v g_u} dy \rightarrow \begin{cases} \frac{\partial z}{\partial x} = \frac{h_u h_v - h_v h_u}{f_u g_v - f_v g_u} \\ \frac{\partial z}{\partial y} = \frac{h_v f_u - h_u f_v}{f_u g_v - f_v g_u} \end{cases}$$

P₂₀ 补充 P₁

$$\begin{cases} z = \lambda f(x, y) \\ F(x, y, z) = 0 \end{cases} \rightarrow \begin{cases} \frac{dz}{d\lambda} = f(x, y) + \lambda f'(x, y) (1 + \frac{dy}{d\lambda}) \\ F_x + F_y \frac{dy}{d\lambda} + F_z \frac{dz}{d\lambda} = 0 \end{cases}$$

$$J = \begin{vmatrix} \lambda f'(x, y) & -1 \\ F_y & F_z \end{vmatrix} = F_z \lambda f'(x, y) + F_y$$

$$\frac{dz}{d\lambda} = - \frac{\begin{vmatrix} \lambda f'(x, y) & -f(x, y) - \lambda f'(x, y) \\ F_y & -F_x \end{vmatrix}}{J} = \frac{(f + \lambda f') F_y - \lambda f' F_x}{F_y + \lambda f' F_z}$$

思考题

P₁ $\begin{cases} x = r \sin \theta \cos \varphi \\ y = r \sin \theta \sin \varphi \\ z = r \cos \theta \end{cases}$

$$\frac{\partial f}{\partial r} = \frac{\partial f}{\partial x} \sin \theta \cos \varphi + \frac{\partial f}{\partial y} \sin \theta \sin \varphi + \frac{\partial f}{\partial z} \cos \theta$$

$$\frac{\partial^2 f}{\partial r^2} = \frac{\partial^2 f}{\partial x^2} (\sin \theta \cos \varphi)^2 + \frac{\partial^2 f}{\partial y^2} (\sin \theta \sin \varphi)^2 + \frac{\partial^2 f}{\partial z^2} \cos^2 \theta = \sin^2 \theta \left(\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} \right) + \cos^2 \theta \frac{\partial^2 f}{\partial z^2}$$

$$\frac{\partial}{\partial r} (r^2 \frac{\partial f}{\partial r}) = 2r \frac{\partial f}{\partial r} + r^2 \frac{\partial^2 f}{\partial r^2}$$

$$\frac{\partial f}{\partial \theta} = \frac{\partial f}{\partial x} r \cos \theta \cos \varphi + \frac{\partial f}{\partial y} r \cos \theta \sin \varphi - \frac{\partial f}{\partial z} r \sin \theta$$

$$\frac{\partial^2 f}{\partial \theta^2} = -\frac{\partial f}{\partial x} r \sin \theta \cos \varphi + \frac{\partial^2 f}{\partial x^2} (r \cos \theta \cos \varphi)^2 - \frac{\partial f}{\partial y} r \sin \theta \sin \varphi + \frac{\partial^2 f}{\partial y^2} (r \cos \theta \sin \varphi)^2 - \frac{\partial f}{\partial z} r \cos \theta - \frac{\partial^2 f}{\partial z^2} (r \sin \theta)^2$$

$$\frac{\partial}{\partial \theta} (\sin \theta \frac{\partial f}{\partial \theta}) = \cos \theta \frac{\partial f}{\partial \theta} + \sin \theta \frac{\partial^2 f}{\partial \theta^2}$$

$$\frac{\partial f}{\partial \varphi} = -\frac{\partial f}{\partial x} r \sin \theta \sin \varphi + \frac{\partial f}{\partial y} r \sin \theta \cos \varphi$$

$$\frac{\partial^2 f}{\partial \varphi^2} = -\frac{\partial f}{\partial x} r \sin \theta \cos \varphi + \frac{\partial^2 f}{\partial x^2} (r \sin \theta \sin \varphi)^2 - \frac{\partial f}{\partial y} r \sin \theta \cos \varphi + \frac{\partial^2 f}{\partial y^2} (r \sin \theta \cos \varphi)^2$$

$$RHS = \frac{1}{2r} \frac{\partial f}{\partial r} + \frac{\partial^2 f}{\partial r^2} + \frac{1}{r^2 \tan \theta} \frac{\partial f}{\partial \theta} + \frac{1}{r^2} \frac{\partial^2 f}{\partial \theta^2} + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 f}{\partial \varphi^2}$$

$$\begin{aligned} &= \left(\frac{\sin \theta \cos \varphi}{2r} + \frac{r \cos \theta \cos \varphi}{r^2 \tan \theta} - \frac{r \sin \theta \cos \varphi}{r^2} - \frac{r \sin \theta \cos \varphi}{r^2 \sin \theta} \right) \frac{\partial f}{\partial x} + \left(\frac{\sin \theta \sin \varphi}{2r} + \frac{r \cos \theta \sin \varphi}{r^2 \tan \theta} - \frac{r \sin \theta \sin \varphi}{r^2} - \frac{r \sin \theta \sin \varphi}{r^2 \sin \theta} \right) \frac{\partial f}{\partial y} \\ &+ \left(\frac{\cos \theta}{2r} - \frac{r \sin \theta}{r^2 \tan \theta} - \frac{r \cos \theta}{r^2} \right) \frac{\partial f}{\partial z} + \left(\sin^2 \theta + \frac{r \cos \theta \cos \varphi^2}{r^2} + \frac{r \sin \theta \sin \varphi^2}{r^2 \sin^2 \theta} \right) \frac{\partial^2 f}{\partial x^2} + \left(\sin^2 \theta + \frac{r \cos \theta \sin \varphi^2}{r^2} + \frac{r \sin \theta \cos \varphi^2}{r^2 \sin^2 \theta} \right) \frac{\partial^2 f}{\partial y^2} \\ &+ \left(\cos^2 \theta + \frac{r \sin \theta^2}{r^2} \right) \frac{\partial^2 f}{\partial z^2} = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2} \quad \square \end{aligned}$$

P_z

$$I) \quad x^i - z^i = (x-z)(x^{i-1} + x^{i-2}z + \dots + xz^{i-2} + z^{i-1})$$

$$p(x) = p(x) - p(z) = \sum_{i=1}^n a_i (x^i - z^i)$$

$$= (x-z) \sum_{i=1}^n a_i (x^{i-1} + x^{i-2}z + \dots + xz^{i-2} + z^{i-1})$$

$$\triangleq q(x) = \sum_{i=1}^n a_i (x^{i-1} + x^{i-2}z + \dots + xz^{i-2} + z^{i-1})$$

则 $q(x)$ 为多项式, 且 $p(x) = (x-z)q(x)$ $\square$.

$$II) \quad p(x) = (x-z)q(x)$$

两边求导 $p'(x) = q(x) + (x-z)q'(x)$

$$p'(z) = q(z) \neq 0 \quad \square.$$

III) $f(a_0, a_1, \dots, a_n; x) = \sum_{i=0}^n a_i x^i$ 是关于 $a_0, \dots, a_n, x$ 的多项式函数, 故在 $\mathbb{R}^{n+2}$ 上可微

$$f(a_0^0, a_1^0, \dots, a_n^0, z^0) = p^0(z^0) = 0, \text{ 零点存在}$$

$$\frac{\partial f}{\partial x} = \sum_{i=1}^n i a_i x^{i-1} = p'(x) \quad \frac{\partial f}{\partial x} \Big|_{a_0^0, a_1^0, \dots, a_n^0, z^0} = (p^0)'(z^0) \neq 0$$

由隐函数定理, 有 z s.t. $f(a_0, a_1, \dots, a_n; z) = 0$

唯一性: $\frac{\partial f}{\partial x} \Big|_{a_0^0, \dots, a_n^0, z^0} \neq 0$ 且连续

故有邻域 U s.t. $\forall (a_0, \dots, a_n) \in U, \frac{\partial f}{\partial x} \Big|_{(a, z(a))} = \sum_{i=1}^n i a_i (z(a))^{i-1} \neq 0$

即 $z(a_0, \dots, a_n)$ 是 $\sum_{i=0}^n a_i x^i$ 的单根 $\rightarrow$ 唯一性得证 $\square$.